

Jacob Pettine
Postdoctoral Associate
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EDUCATION & EXPERIENCE

Postdoctoral Associate | Massachusetts Institute of Technology 2024–Present
Department of Physics, Gedik Group

Director's Postdoctoral Fellow | Los Alamos National Laboratory 2021–2024
CINT, Laboratory for Ultrafast Materials and Optical Sciences

Instructor | CU Boulder Physics Department 2021
Upper Division Quantum Mechanics (Co-Instructed with Charles Rogers)

NSF Graduate Research Fellow | CU Boulder/JILA 2014–2020
Ph.D. in Physics

- *Advisor:* David J. Nesbitt
- *Thesis topic:* Probing the femtosecond dynamics of hot electrons in metal nanoparticles via time- and angle-resolved photoemission spectroscopy

Undergraduate Research Assistant | CU Boulder/JILA 2010–2013
B.S. in Engineering Physics, GPA: 3.9/4.0

- *Advisor:* Dana Z. Anderson
- *Research topics:* Self-adaptive holographic optical filters and photonic crystal fiber waveguides for cold atoms

GRANTS & AWARDS

- Publication Prize in Experimental Science | LANL 2025
- Lindau Nobel Laureate Meeting Fellowship | UC/LANL 2024
- Exploratory Research Grant (Co-PI) | LANL LDRD 2022
 - Co-PI on \$1M fund for exploring ultrafast, nanoscale dynamics in hybrid low-dimensional materials (~10% acceptance rate)
- Institute for Materials Science Grant (PI) | LANL LDRD 2022
 - PI on \$50k fund for exploring non-Hermitian nanophotonics
- Director's Postdoctoral Fellowship | LANL 2021
- CU Physics Award for Postdoc Teaching Excellence | CU Boulder 2021

- Beverly Sears Graduate Research Grant | CU Boulder 2020
- APS Harry Lustig Award Finalist | APS 4CS 2020
- GRC Poster Award | Noble Metals GRC 2018
- NSF Graduate Research Fellowship | NSF 2015

PUBLICATIONS

* corresponding, [†]equal contribution

In Press

- A von Hoegen, T Tai, CJ Allington, M Yeung, J Pettine, M Michael, E Viñas Boström, X Cui, K Torres, AE Kossak, B Lee, GSD Beach, A Rubio, P Kim, N Gedik, “Visualizing a Terahertz Superfluid Plasmon in a Two-Dimensional Superconductor,” *Nature* (Accepted), *arXiv:2506.08204*

Reviews and perspectives

- J Pettine^{*}, HT Chen^{*}, “Making and Breaking Terahertz Waves with Fluids,” *Photonics Insights* **2**, C05 (2023) [Commentary]
- J Pettine[†], N Sirica[†], P Padmanabhan[†], AJ Taylor, RP Prasankumar, HT Chen^{*}, “Ultrafast Terahertz Emission from Emerging Symmetry-Broken Materials,” *Light: Sci. & Appl.* **12**, 133 (2023) [Review]
- J Pettine^{*}, DJ Nesbitt^{*}, “Emerging Methods for Controlling Excitation and Emission Distributions of Hot Carriers in Nanoplasmonic Systems,” *J. Phys. Chem. C* **126**, 14767-14780 (2022) [Perspective]

Research articles

- J Pettine^{*}, P Padmanabhan, T Shi, L Gingras, L McClintock, CC Chang, KWC Kwock, L Yuan, Y Huang, J Nogan, JK Baldwin, P Adel, R Holzwarth, AK Azad, F Ronning, AJ Taylor, RP Prasankumar, SZ Lin, HT Chen^{*}, “Light-Driven Nanoscale Vectorial Currents,” *Nature* **626**, 984 (2024)
- J Pettine^{*}, P Maioli, F Vallee, N Del Fatti, DJ Nesbitt^{*}, “Energy-Resolved Femtosecond Hot Electron Dynamics in Single Plasmonic Nanoparticles,” *ACS Nano* **17**, 10721-10732 (2023)
- F Medeghini, J Pettine, SM Meyer, CJ Murphy, DJ Nesbitt^{*}, “Regulating and Directionally Controlling Electron Emission from Gold Nanorods with Silica Coatings,” *Nano Lett.* **22**, 644-651 (2022)
- J Pettine, SM Meyer, F Medeghini, CJ Murphy, DJ Nesbitt^{*}, “Controlling the Spatial and Momentum Distributions of Plasmonic Carriers: Volume vs Surface Effects,” *ACS Nano* **15**, 1566-1578 (2021)
- SM Meyer, J Pettine, DJ Nesbitt, CJ Murphy^{*}, “Size Effects in Gold Nanorod Light-to-Heat Conversion under Femtosecond Illumination,” *J. Phys. Chem. C* **125**, 16268-16278 (2021)
- J Pettine, P Choo, F Medeghini, TW Odom^{*}, DJ Nesbitt^{*}, “Plasmonic Nanostar Photocathodes for Optically-Controlled Directional Currents,” *Nat. Commun.* **11**, 1367 (2020)
- J Pettine, M Zhu, DZ Anderson^{*}, “Self-Adaptive Amplified Spontaneous Emission Suppression with a Photorefractive Two-Beam Coupling Filter,” *Opt. Lett.* **45**, 6258-6261 (2020)
- J Pettine, A Marton Menendez, DJ Nesbitt^{*}, “Continuous Angular Control over Anisotropic Photoemission from Isotropic Gold Nanoshells,” *J. Chem. Phys.* **153**, 101101 (2020)

- J Pettine, A Grubisic, DJ Nesbitt*, “Polarization-Controlled Directional Multiphoton Photoemission from Hot Spots on Single Au Nanoshells,” *J. Phys. Chem. C* **122**, 14805-14813 (2018)
- J Pettine, A Grubisic, DJ Nesbitt*, “Angle- and Momentum-Resolved Velocity Map Imaging Studies of Thin Au Film and Single Supported Au Nanoshells,” *J. Phys. Chem. C* **122**, 3970-3984 (2018)

TALKS & POSTERS

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| 07/2025 | <i>Time-Resolved Terahertz Evidence for Unconventional Heavy Fermions in a Kagome Metal</i> , Conference on Strongly Correlated Electron Systems, Montreal, Canada
(Contributed Talk) |
| 06/2025 | <i>Tuning the Properties of Quantum Materials through Nanostructured Symmetries</i> , EPiQS Postdoctoral Symposium, Pasadena, CA
(Contributed Talk) |
| 05/2025 | <i>Bespoke Symmetries and Tailored Flows: Controlling Electrical Currents with Light</i> , Postdoctoral Prize in Experimental Science, Los Alamos National Laboratory, NM
(Awarded Talk) |
| 04/2025 | <i>Tuning the Properties of Quantum Materials through Nanostructured Symmetries</i> , Condensed Matter Physics Seminar, University of Connecticut, Storrs, CT
(Invited Talk) |
| 11/2024 | <i>Nanoscale Vectorial Photocurrents in Symmetry-Broken Metasurfaces</i> , FAU Chair of Laser Physics Autumn Retreat, Erlangen, Germany
(Invited Talk) |
| 02/2024 | <i>Light-Driven Currents in Emerging Materials</i> , Ultrafast Phenomena in Cooperative Systems, Lucca, Italy
(Poster) |
| 10/2023 | <i>Light-Based Nanoscale Current Control in Symmetry-Broken Optoelectronic Metasurfaces</i> , Frontiers in Optics + Laser Science, Tacoma, WA
(Invited Talk) |
| 09/2023 | <i>Vectorial Currents and Broadband Terahertz Vector Beams with Optoelectronic Metasurfaces</i> , IRMMW-THz, Montreal, Canada
(Invited Keynote Talk) |
| 03/2023 | <i>Ultrafast Vectorial Currents in Nanoscale Symmetry-Controlled Optoelectronic Metasurfaces</i> , APS March Meeting, Las Vegas, NV
(Contributed Talk) |
| 10/2022 | <i>Nanoscale Vectorial Current Control in Symmetry-Broken Optoelectronic Metasurfaces</i> , Ultrafast Phenomena in Cooperative Systems GRC, Ventura, CA
(Poster) |
| 05/2022 | <i>Ultrafast Nanoscale Photocurrent Control and Terahertz Emission in Hybrid Plasmonic-2D Material Systems</i> , CLEO (Virtual)
(Contributed Talk) |

03/2022	<i>Bulk-Like Femtosecond Dynamics of Ballistic Hot Carriers in Single Plasmonic Gold Nanoparticles</i> , ACS Spring Meeting (Virtual) (Contributed Talk)
03/2021	<i>Optically Controlling Femtosecond Hot Electron Spatial and Momentum Distributions in Nanoplasmonic Systems</i> , APS March Meeting (Virtual) (Contributed Talk)
10/2020	<i>Optically Controlling Femtosecond Hot Electron Currents in Nanoplasmonic Systems</i> APS 4CS Meeting, Harry Lustig Award Session (Virtual) (Awarded Talk)
03/2020	<i>Plasmonic Nanostar Photocathodes for Optically-Controlled Directional Currents</i> , University of Colorado March Physics Symposium, Boulder, CO (Contributed Talk)
10/2018	<i>Nanoplasmonics: Controlling Photons and Photocurrents at the Femto-Nanoscale</i> , Colorado Photonics Industry Association Meeting, Boulder, CO (Poster)
07/2018	<i>Real Stories from Reciprocal Space: Angle-Resolved Photoemission Studies of Single Plasmonic Nanoparticles</i> , Noble Metals GRC, South Hadley, MA (Awarded Talk)
07/2016	<i>Ultrafast Energy- and Angle-Resolved Photoemission Studies of Single Supported Plasmonic Nanoparticles</i> , Plasmonics and Nanophotonics GRC, Newry, ME (Poster)
06/2014	<i>Coherent Multiphoton Photoelectron Emission from Individual Supported Au Nanorods: Angle- and Energy-Resolved Photoelectron Distribution Measurements via Velocity Map Imaging</i> , Excited State Processes, Santa Fe, NM (Poster)

TEACHING & OUTREACH

- MIT UROP mentorship (2024, 2025)
 - Mentoring three students; nominated for 2025 Outstanding UROP Mentor
- High school science showcase (2025)
 - Science demonstrations at Cambridge Rindge and Latin school
- Tour of Los Alamos Neutron Science facility (2023)
 - Organized tour of LANSCE for a group of high school physics teachers
- Co-Instructor, CU Boulder PHYS-4410, Quantum Mechanics II (Spring 2021)
- CU Prime (2015, 2016)
 - Monthly meetings and mentorship of two undergraduate physics students
- Partnership for Informal Science Education in the Community (2014, 2015)
 - Afterschool program for K–12 students in underrepresented communities
- CU Wizards (2016, 2018, 2020)
 - Assisted with community science presentations oriented toward young audience